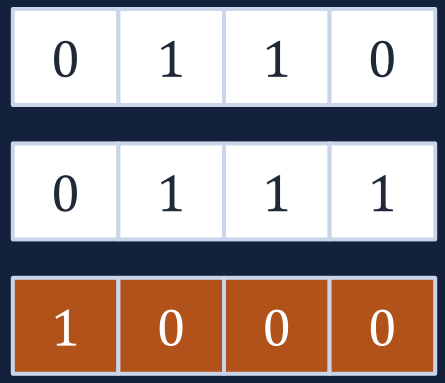
ENES 246 · DIGITAL LOGIC DESIGN

Lecture 22

Synchronous Counters

Synchronous binary counters · up-down counters · parallel load · BCD and arbitrary sequences · ring and Johnson counters



every bit on the same edge

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 6, §6.4–6.5

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Today’s roadmap

1 §6.4

Binary counters

One clock for every bit

2 §6.4

Up-down, load

Count either way; preset

3 §6.5

BCD, arbitrary

Any sequence; unused states

4 §6.5

Ring, Johnson

Shift-register counters

By the end of class you can…

- Derive the T inputs of a synchronous binary counter
- Add up-down control and parallel load
- Design a BCD counter and check its unused states
- Design a counter for an arbitrary sequence
- Trace ring and Johnson counters and decode their states

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Agenda for today's session



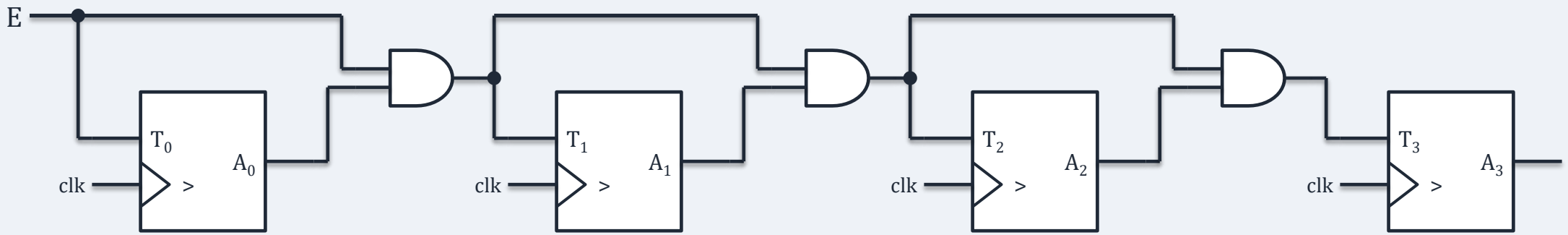
Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome; Homework 10 collected	—
0:05 – 0:25	Synchronous binary counters	Example 1
0:25 – 0:40	Up-down counters	Example 2
0:40 – 0:50	Break	
0:50 – 1:15	Parallel load; the BCD counter	Example 3
1:15 – 1:40	Arbitrary sequences; ring and Johnson	Example 4 · Your turn
1:40 – 1:50	Concept check and wrap-up	Exam 3 preview

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Example 1 — A synchronous counter

§6.4

EXAMPLE Design a 4-bit synchronous binary up-counter with T flip-flops and a count enable E.



1. Binary counting: a bit toggles when every lower bit is 1
2. T flip-flops: $T = 1$ means toggle on the next edge
3. Bit 0 toggles on every enabled clock: $T_0 = E$
4. $T_1 = EA_0,\ T_2 = EA_0A_1,\ T_3 = EA_0A_1A_2$
5. Each AND adds one bit to the previous enable
6. One shared clock: no false counts

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Example 2 — An up-down counter

EXAMPLE

3-bit counter with inputs Up and Down (Up wins if both are 1). Find the T inputs.

Up	Down	Action
1	X	count up
0	1	count down
0	0	hold

Down-count rule

Counting down, a bit toggles when every lower bit is 0: the borrow.

1. Enables: $U = Up;\ D_n = Up'\,Down$ (Up wins)
2. Bit 0 toggles on every up or down count: $T_0 = U + D_n$
3. Up: bit 1 toggles when $A_0 = 1 \rightarrow UA_0$
4. Down: bit 1 toggles when $A_0 = 0$ (a borrow) $\rightarrow D_n A_0'$
5. $$T_1 = UA_0 + D_n A_0'$$
6. Same rule for bit 2: $T_2 = UA_0A_1 + D_n A_0' A_1'$
7. Checked on all 32 cases of Up, Down and state ✓

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Break

10 minutes — we resume at the 0:50 mark

Next: counters with load, BCD and other sequences

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Counters with parallel load

§6.4

Control inputs

Clear: force 0000. Load: copy the inputs $I_3 \ldots I_0$. Count: add 1. Otherwise hold.

Each stage chooses its next value with a multiplexer (as in Lecture 21), not by gating the clock.

Build any modulus

Load 0000 when the count reaches $N - 1$: a mod-N counter.

Mod 10: detect 1001 with $Q_3 Q_0$ and assert Load with inputs 0000.

Clear	Load	Count	Next
1	X	X	0000
0	1	X	$I_3 I_2 I_1 I_0$
0	0	1	A + 1
0	0	0	A (hold)

A synchronous clear or load acts at the next clock edge.

An asynchronous clear acts immediately and can cause a glitch of the unwanted state.

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Example 3 — The BCD counter

§6.5

EXAMPLE Design a synchronous BCD counter (0–9) with T flip-flops and output $y = 1$ at count 9. Check the unused states.

$Q_8Q_4Q_2Q_1$	next	$T_8T_4T_2T_1$
0000	0001	0001
0001	0010	0011
0010	0011	0001
0011	0100	0111
0100	0101	0001
0101	0110	0011
0110	0111	0001
0111	1000	1111
1000	1001	0001
1001	0000	1001

1. Maps with 10–15 as don't-cares give:

2. $T_{Q1} = 1,\ T_{Q2} = Q_8' Q_1,\ T_{Q4} = Q_2 Q_1$

3. $T_{Q8} = Q_8 Q_1 + Q_4 Q_2 Q_1,\ y = Q_8 Q_1$

4. Unused states, from these equations:

from	10	11	12	13	14	15
to	11	6	13	4	15	2

Every unused state reaches the valid count within two clocks: self-correcting.

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Example 3 — Step by step

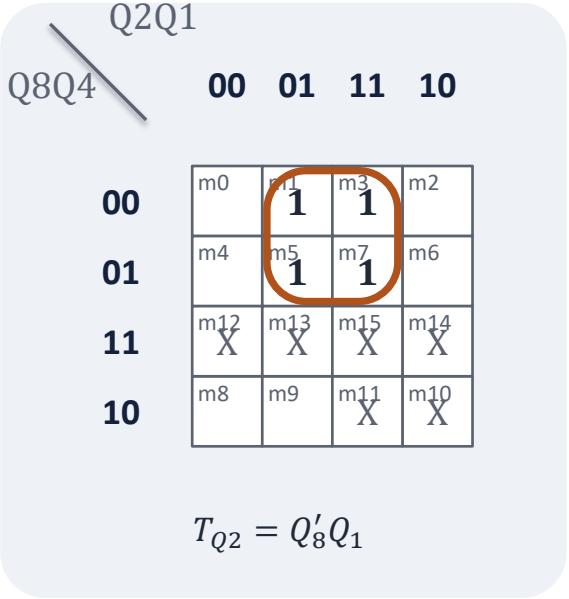
§6.5

1. For each count, T = present $\oplus$ next, bit by bit
2. Map each T column; 10–15 are don't-cares
3. T_{Q1} is 1 on every row → $T_{Q1} = 1$
4. $T_{Q2} = 1$ on 0001, 0011, 0101, 0111 → $Q_8' Q_1$
5. $T_{Q4} = 1$ on 0011, 0111 → $Q_2 Q_1$
6. $T_{Q8} = 1$ on 0111, 1001 → $Q_8 Q_1 + Q_4 Q_2 Q_1$
7. $y = 1$ only at 1001; with X → $y = Q_8 Q_1$
8. Unused states: put 10–15 into the equations
9. e.g. 10 (1010): only $T_{Q1} = 1$ → 1011 = 11
10. Each unused state reaches a valid count in ≤ 2 clocks

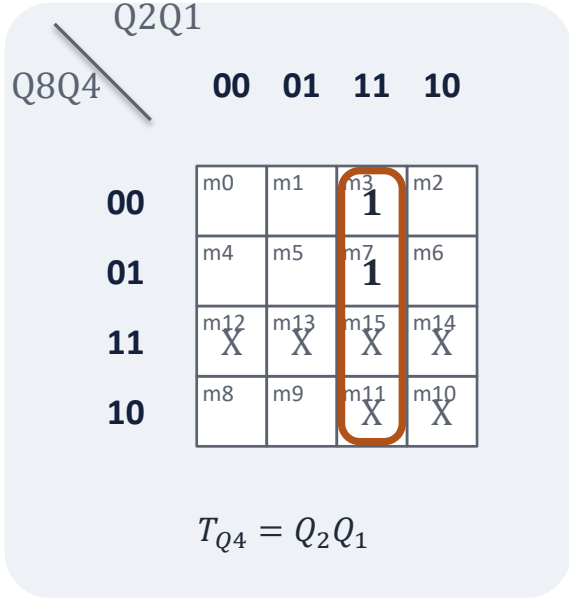
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Example 3 — The K-maps

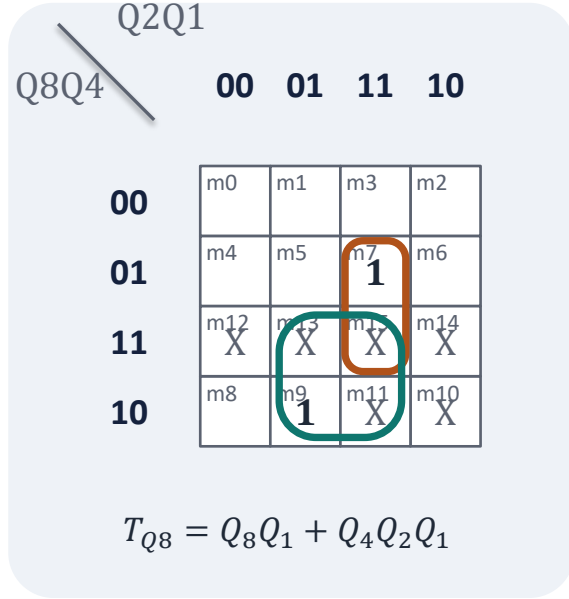
§6.5



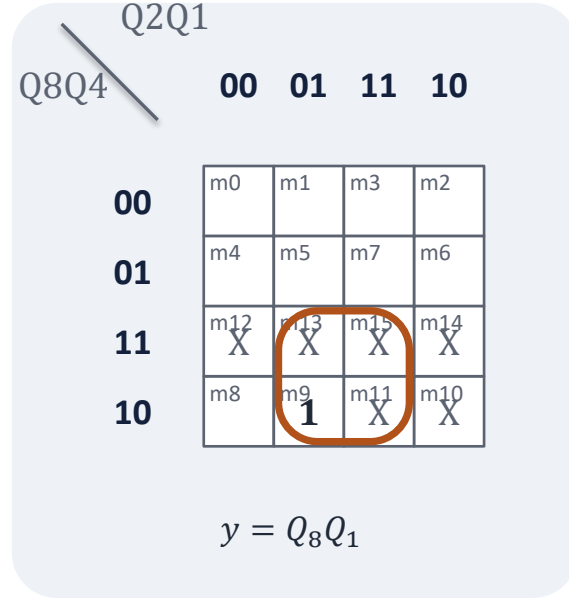
$T_{Q2} = Q_8' Q_1$



$T_{Q4} = Q_2 Q_1$



$T_{Q8} = Q_8 Q_1 + Q_4 Q_2 Q_1$



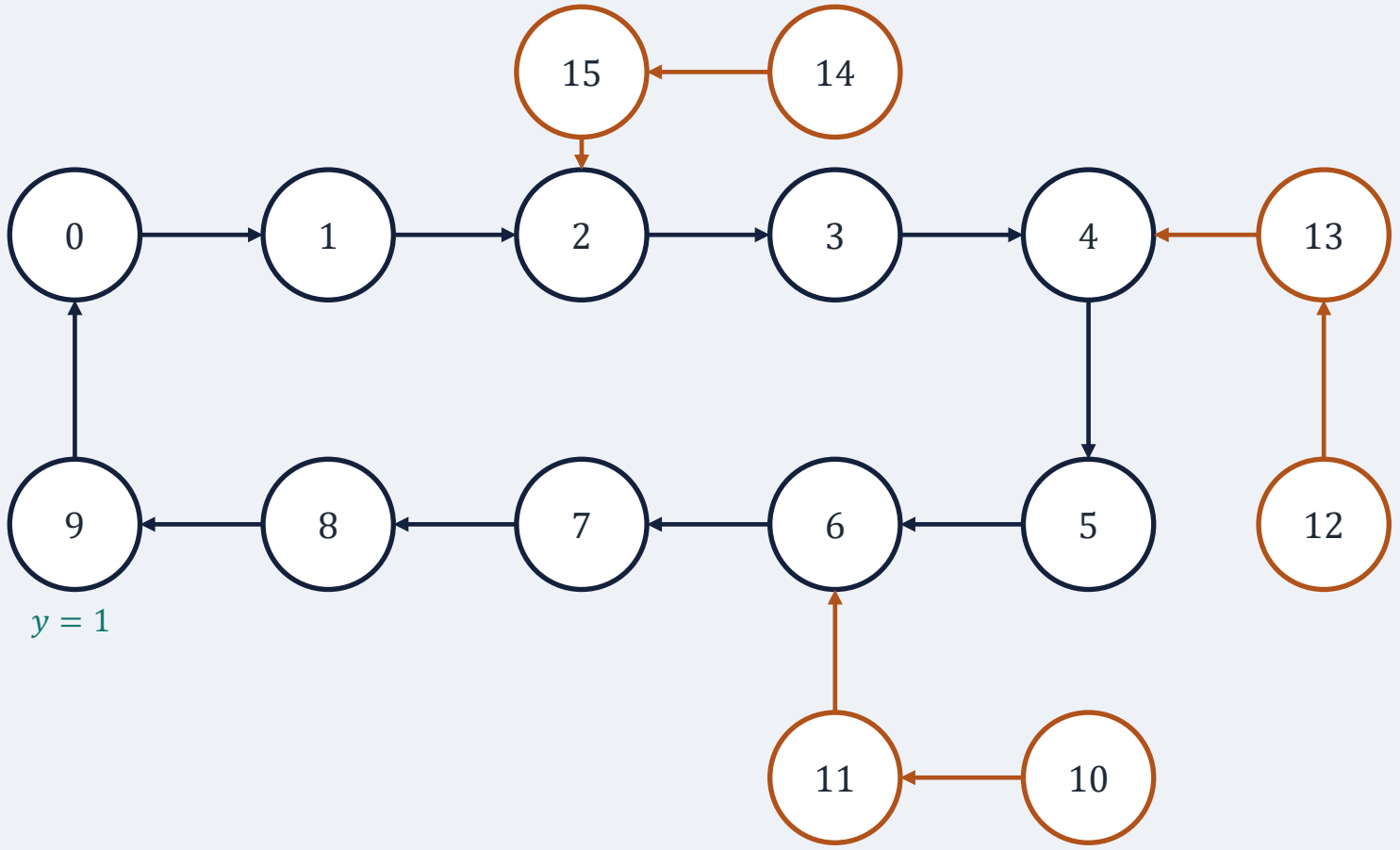
$y = Q_8 Q_1$

1. 1 at 1, 3, 5, 7: Q_2 flips there
2. Quad m1, m3, m5, m7 → $Q_8' Q_1$
3. At 9, Q_2 must not flip: no X helps
4. 1 at 3, 7: Q_4 flips there
5. X at 11, 15 completes a quad
6. Quad m3, m7, m11, m15 → $Q_2 Q_1$
7. 1 at 7 and 9: Q_8 flips there
8. m7 + X at m15 → $Q_4 Q_2 Q_1$
9. m9 + X at 11, 13, 15 → $Q_8 Q_1$
10. $y = 1$ only at count 9
11. m9 + X at 11, 13, 15 → $Q_8 Q_1$
12. $T_{Q1} = 1$: Q_1 flips on every count (no map)

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Example 3 — All 16 states

§6.5



Reading the diagram

Navy: the count 0 → 9 → 0.

Copper: the unused codes 10–15.

States are in decimal: 6 is $Q_8Q_4Q_2Q_1 = 0110$.

Their arrows come from the same T equations:

10 → 11 → 6

12 → 13 → 4

14 → 15 → 2

Every unused state reaches the count within two clocks: the counter is self-correcting.

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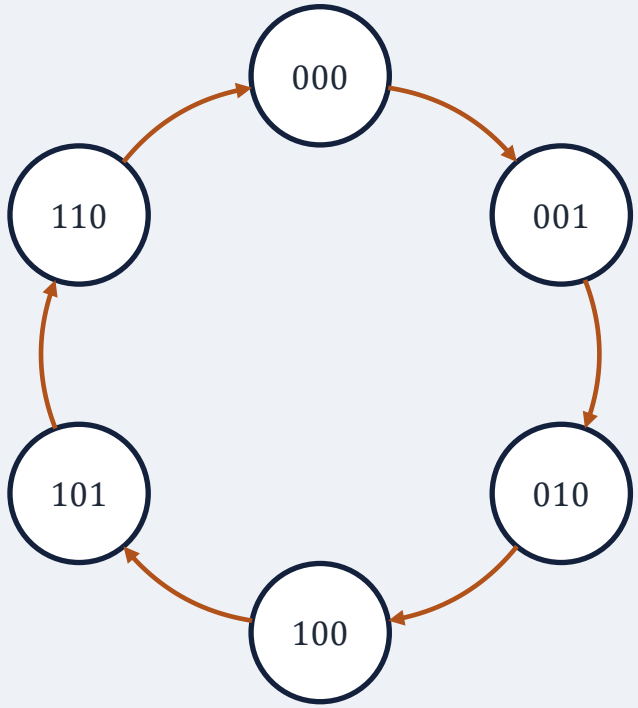
Example 4 — An arbitrary sequence

§6.5

EXAMPLE

Design a counter with JK flip-flops that repeats 0, 1, 2, 4, 5, 6. States 3 and 7 are unused.

ABC	next	JA KA	JB KB	JC KC
000	001	0 X	0 X	1 X
001	010	0 X	1 X	X 1
010	100	1 X	X 1	0 X
100	101	X 0	0 X	1 X
101	110	X 0	1 X	X 1
110	000	X 1	X 1	0 X



1. $J_A = K_A = B$
2. $J_B = C, K_B = 1$
3. $J_C = B', K_C = 1$
4. Unused: 011 → 100, 111 → 000
5. Both recover ✓

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Example 4 — Step by step

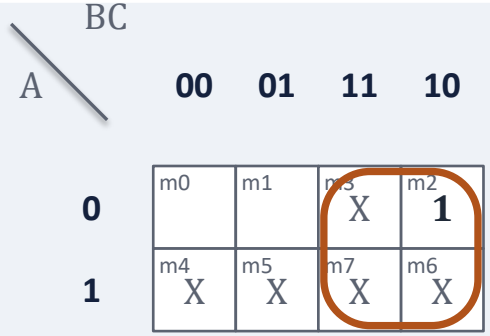
§6.5

1. Next state for each present state; 3 and 7 are don't-cares
2. For each bit, J, K from the excitation table (bit → next bit)
3. $J_A = 1$ only at 010 (→ 100); X where $A = 1 \rightarrow J_A = B$
4. $K_A = 1$ at 110 (→ 000) → $K_A = B$
5. $J_B = 1$ at 001, 101 → $J_B = C$; every K_B is 1 or X → $K_B = 1$
6. $J_C = 1$ at 000, 100 → $J_C = B'$; $K_C = 1$
7. Unused 011: A: $JK = 11$ → 1; B: 11 → 0; C: 01 → 0
8. So 011 → **100**
9. Unused 111: A toggles → 0; B toggles → 0; C resets → 0
10. So 111 → **000**: both recover ✓

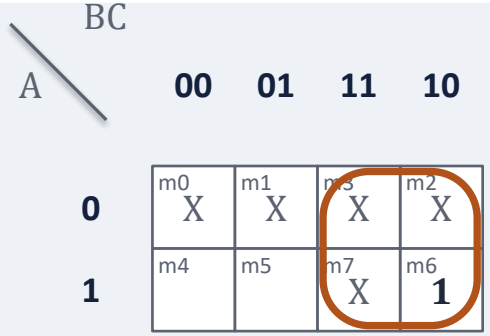
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Example 4 — The six K-maps

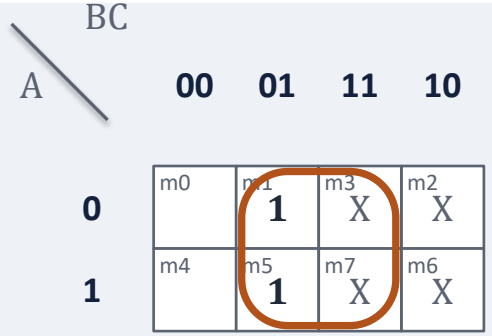
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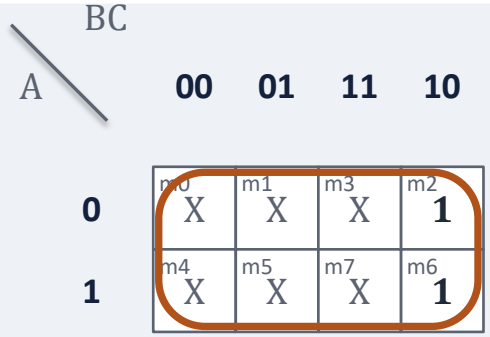
$J_A = B$



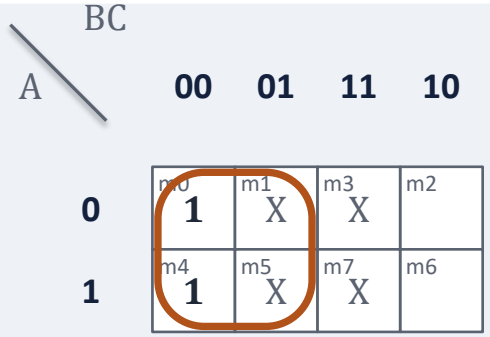
$K_A = B$



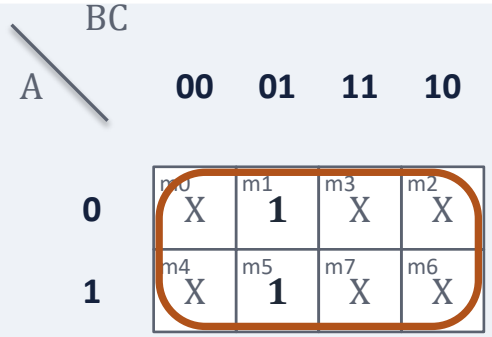
$J_B = C$



$K_B = 1$



$J_C = B'$



$K_C = 1$

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Ring and Johnson counters

§6.5

Ring counter: $Q_0 \rightarrow D_3$

Clock	$Q_3Q_2Q_1Q_0$
0	1000
1	0100
2	0010
3	0001

4 flip-flops, 4 states, one hot bit: no decoder needed.

Johnson counter: $Q_0' \rightarrow D_3$

Clock	$Q_3Q_2Q_1Q_0$	Decode
0	0000	A′E′
1	1000	AB′
2	1100	BC′
3	1110	CE′
4	1111	AE
5	0111	A′B
6	0011	B′C
7	0001	C′E

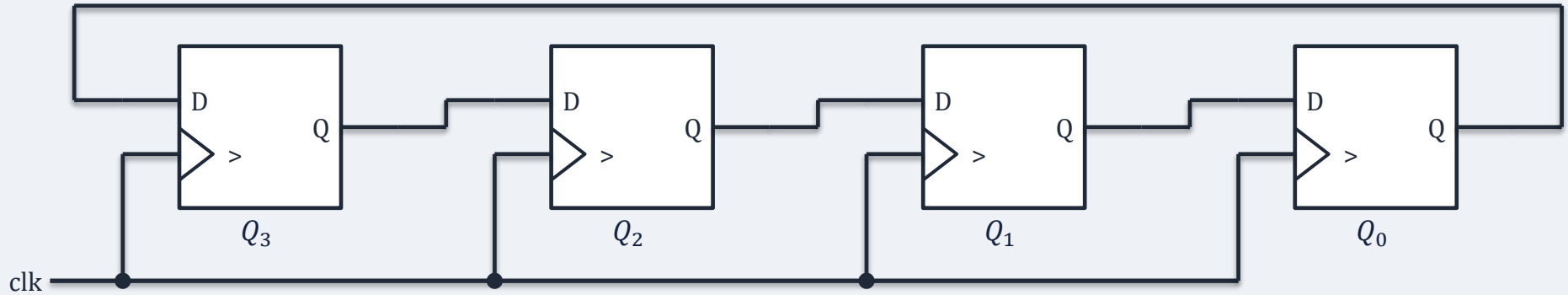
4 flip-flops, 8 states; each state decoded by one 2-input AND.

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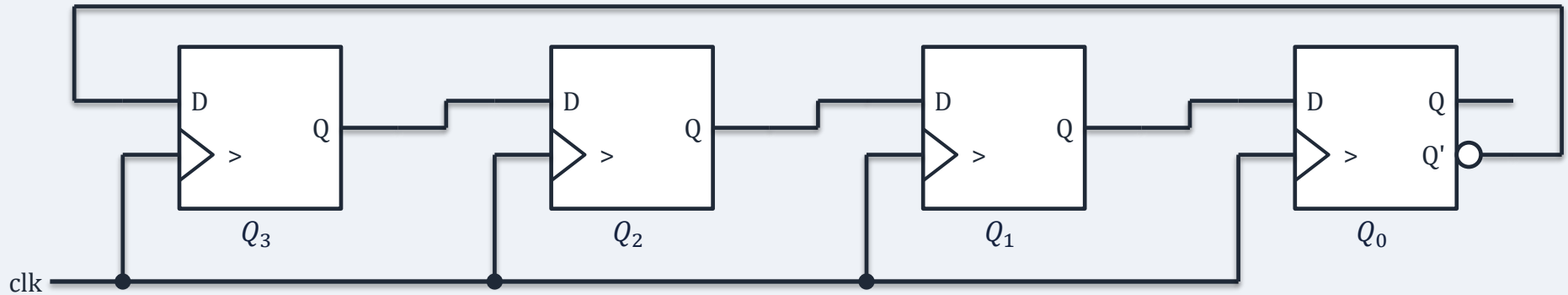
Ring and Johnson circuits

§6.5

Ring: Q_0 feeds back



Johnson: Q_0' feeds back



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Your turn — Up-down with D flip-flops

§6.4

YOUR TURN

≈ 12 min

A 2-bit counter A_1A_0 counts up when $U = 1$ and down when $U = 0$.

(a) Write the state table (U, A_1, A_0 → next state).

(b) Find D_1 and D_0 with K-maps. Can you write D_1 with XOR gates?

(c) How many states does a 5-flip-flop Johnson counter have?

Map variables in the order U, A_1, A_0.

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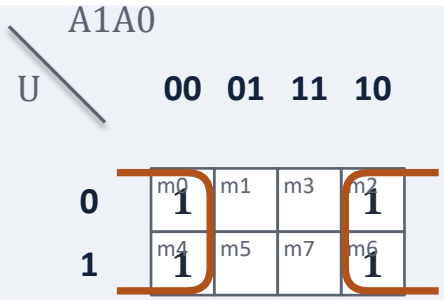
Your turn — Solution

SOLUTION

U	A_1A_0	next
0	00	11
0	01	00
0	10	01
0	11	10
1	00	01
1	01	10
1	10	11
1	11	00



$$D_1 = A_1 \oplus A_0 \oplus U'$$



$$D_0 = A_0'$$

1. (a) $U = 1$: next = present + 1; $U = 0$: present − 1
2. (b) D flip-flops: D = next-state bit
3. D_0 map: quad $A_0 = 0 \rightarrow D_0 = A_0'$
4. D_1 map: 1s at m0, m3, m5, m6
5. A checkerboard: no two 1s adjacent → XOR form
6. $$D_1 = A_1 \oplus A_0 \oplus U'$$
7. Two XOR gates and an inverter
8. (c) Johnson: $2n =$ **10 states**

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Concept check

1. Why does a synchronous counter show no false counts?

2. When does bit i toggle in an up-counter? In a down-counter?

3. Why check the unused states of a BCD counter?

4. How many states do ring and Johnson counters with n flip-flops have?

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Key takeaways

Binary

$$T_i = E\,A_0 \cdots A_{i-1}$$

Down

toggle when lower bits are 0

Mod N

load 0 at count $N - 1$

BCD

$$T_{Q8} = Q_8 Q_1 + Q_4 Q_2 Q_1$$

Arbitrary

excitation table + X's

Ring / Johnson

n / $2n$ states

Same clock, every bit: synchronous counters count cleanly.

Homework #11 due at the start of Lecture 28

This afternoon's lab: Lab Hp4 — Calculator 4: BCD to seven-segment display

Next: review for Exam 3

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